

KRASNOYARSK, Russian Federation

WMO: 295740

Lat: 56.000N

Lon: 92.883E

Elev: 909

StdP: 14.22

Time Zone: 7.00 (E07)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.6	-23.9	-33.0	0.9	-28.3	-28.1	1.2	-23.6	28.2	12.0	26.0	12.1	2.8	290	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	83.1	65.0	79.9	63.7	76.6	62.3	68.0	78.2	66.2	75.7	64.5	73.3	6.4	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	94.1	72.3	62.6	87.9	70.4	60.9	82.7	69.0	33.0	78.2	31.5	75.9	30.2	73.8	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.7	18.8	15.8	-31.6	88.8	7.1	3.7	-36.7	91.5	-40.9	93.7	-44.9	95.7	-50.1	98.4		
(5)			-30.7	71.8	7.4	3.6	-36.1	74.4	-40.5	76.5	-44.6	78.5	-50.0	81.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.5	5.7	7.8	20.0	34.3	49.1	59.3	65.2	60.1	47.8	35.6	18.9	8.4	
	DBStd	23.45	15.02	13.40	11.04	10.06	9.47	7.62	5.70	5.35	6.91	8.80	12.82	14.46	
	HDD50	6907	1372	1182	929	474	136	14	0	2	123	452	934	1289	
	HDD65	11257	1837	1602	1394	921	500	199	69	167	517	913	1384	1754	
	CDD50	1249	0	0	0	4	107	292	471	315	56	4	0	0	
	CDD65	122	0	0	0	0	6	28	74	14	0	0	0	0	
	CDH74	1227	0	0	0	0	101	332	654	133	7	0	0	0	
	CDH80	247	0	0	0	0	23	61	149	14	0	0	0	0	
Wind	WSAvg	6.3	6.0	5.9	6.4	8.3	7.3	5.5	4.5	4.5	5.0	7.3	7.7	7.1	
Precipitation	PrecAvg	19.5	0.7	0.6	0.7	1.2	1.8	2.4	3.0	2.9	1.8	1.7	1.6	1.1	
	PrecMax	26.2	1.4	1.6	1.9	2.9	3.9	5.8	7.7	5.9	4.4	5.6	4.7	2.8	
	PrecMin	12.7	0.1	0.1	0.1	0.1	0.2	0.7	0.3	0.6	0.5	0.2	0.2	0.2	
	PrecStd	3.3	0.3	0.3	0.4	0.6	0.9	1.1	1.3	1.3	0.9	0.9	0.9	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.7	38.6	49.7	64.8	83.3	85.3	88.0	81.8	74.3	63.4	43.1	37.9	
		MCWB	29.3	31.4	40.1	47.7	58.5	65.0	68.6	65.5	59.0	48.7	37.2	31.3	
	2%	DB	31.4	32.9	42.7	59.2	76.5	81.8	84.2	77.8	68.7	57.0	39.9	33.2	
		MCWB	27.1	27.2	34.7	44.5	56.1	63.5	66.7	62.9	57.0	46.1	33.8	28.0	
	5%	DB	28.6	29.3	38.7	53.9	71.6	78.3	81.0	74.8	63.7	52.6	37.0	29.5	
		MCWB	25.1	24.9	32.3	42.8	54.0	61.9	66.0	61.7	54.2	44.3	32.2	25.7	
	10%	DB	25.8	25.2	35.0	49.1	66.4	74.2	77.6	71.6	59.9	48.1	34.4	26.5	
		MCWB	22.6	21.6	30.2	40.1	51.5	59.8	64.8	60.6	52.2	41.3	30.5	23.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.6	31.9	40.1	50.3	61.0	68.5	72.6	67.5	61.7	50.6	39.8	32.8
			MCDB	35.0	37.8	49.1	58.4	76.3	82.1	82.0	75.0	70.2	60.3	41.8	36.4
2%		WB	27.6	27.7	35.3	46.4	57.8	65.9	69.1	65.5	57.6	47.6	34.9	28.5	
		MCDB	31.0	32.9	41.4	56.8	72.8	77.7	79.2	73.9	65.6	54.7	38.0	33.0	
5%		WB	25.2	25.0	32.8	43.2	55.2	63.7	67.5	63.7	55.3	45.2	32.9	26.0	
		MCDB	28.6	29.1	37.9	53.5	68.6	74.3	77.3	71.7	62.6	52.1	36.4	29.4	
10%		WB	22.6	21.7	30.6	40.4	53.1	61.5	65.8	61.9	53.1	42.2	30.6	23.5	
		MCDB	25.9	25.1	35.0	48.3	64.7	71.2	74.8	69.2	58.8	47.4	34.2	26.4	
Mean Daily Temperature Range	5% DB	MDBR	14.3	16.0	17.1	16.8	19.2	19.9	18.6	18.2	17.4	13.1	12.0	13.1	
		MCDBR	14.0	16.9	19.7	24.7	28.4	26.8	24.7	24.4	25.5	21.1	12.9	16.0	
		MCWBR	11.4	13.4	13.3	14.0	13.6	11.7	10.5	12.1	14.4	12.8	10.0	12.8	
	5% WB	MCDBR	13.7	16.3	18.5	23.1	24.4	22.5	20.5	19.6	22.3	19.0	13.8	15.7	
		MCWBR	11.5	13.3	13.1	14.0	12.4	11.1	10.3	11.7	13.4	12.7	11.7	13.1	
Clear-Sky Solar Irradiance	taub		0.281	0.314	0.361	0.394	0.411	0.407	0.429	0.410	0.368	0.349	0.316	0.283	
	taud		2.031	2.014	1.979	2.141	2.198	2.307	2.258	2.277	2.346	2.246	2.072	1.977	
	Ebn at Noon		186	227	246	259	263	266	256	252	248	217	168	147	
	Edh at Noon		21	31	42	42	42	38	39	36	30	25	20	17	
All-Sky Solar Radiation	RadAvg		255	569	996	1361	1563	1883	1676	1346	867	460	230	151	
	RadStd		28	36	76	88	167	156	113	119	114	37	31	19	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (19 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page